

Name: MODEL SOLUTIONS.

Student Number:

Please answer ALL questions.

This test is worth 25% of your overall class grade.
Each problem statement has the associated mark value stated.

Do not turn over your cover sheet to start until you are told to begin. Extra blank pages are given for each problem with what should be enough room to answer each question, especially if you use both sides. However, if you need extra paper, please just ask and it will be provided.

If you separate the pages, ensure that they are stapled together again when handed in.

There is a total of 40 marks (21 marks for Question 1 and 19 marks for Question 2)

90 minutes total time allowed (plus an additional 5-10 minutes to upload your code).

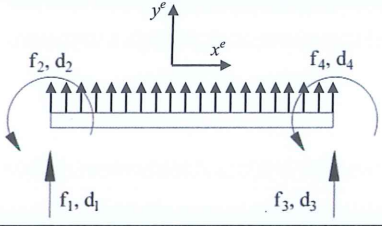
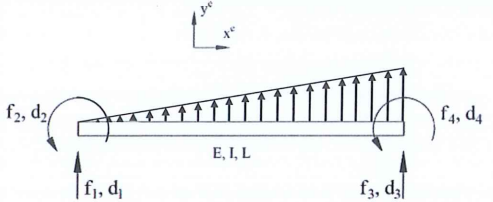
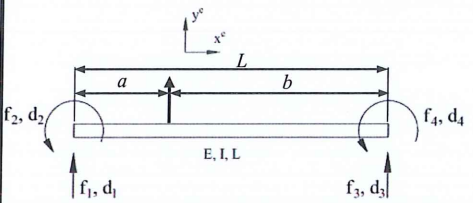
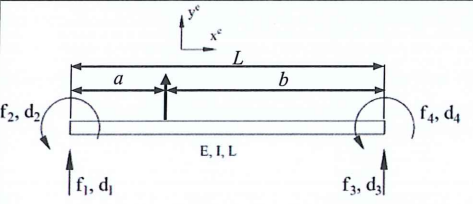
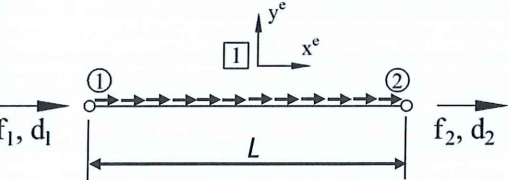
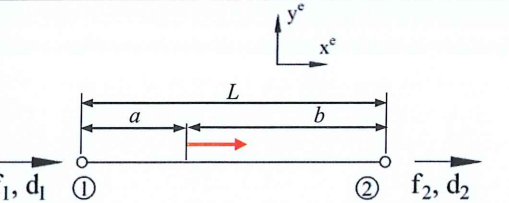
Circle or box all final answers! You may use pencil but make sure to highlight and/or ink final answers.

Label each part of each question, include appropriate units, and an appropriate number of significant figures, for all answers.

Some equations for equivalent nodal loading are presented on the next page for your reference.

At the completion of the test, you are asked to upload the code that you used to complete the test via the link on the course Ako | Learn page. While there are no marks allocated for your code, having access to your code helps with the awarding of partial credit when final answers are incorrect.

The equivalent nodal loading definitions (in element co-ordinates) are provided for reference.

Equivalent nodal loading definition for a Uniformly Distributed Load (UDL)		
Beam Element Formulation	Frame Element Formulation	
	$f_{eq,UDL}^e = \begin{Bmatrix} \bar{w}L/2 \\ \bar{w}L^2/12 \\ \bar{w}L/2 \\ -\bar{w}L^2/12 \end{Bmatrix}$	$f_{eq,UDL\{6 \times 1\}}^e = \begin{Bmatrix} 0 \\ \bar{w}L/2 \\ \bar{w}L^2/12 \\ 0 \\ \bar{w}L/2 \\ -\bar{w}L^2/12 \end{Bmatrix}$
Equivalent nodal loading definition for a Linearly Varying Load (LVL)		
Beam Element Formulation	Frame Element Formulation	
	$f_{eq,LVL}^e = \begin{Bmatrix} 3\bar{w}L/20 \\ \bar{w}L^2/30 \\ 7\bar{w}L/20 \\ -\bar{w}L^2/20 \end{Bmatrix}$	$f_{eq,LVL\{6 \times 1\}}^e = \begin{Bmatrix} 0 \\ 3\bar{w}L/20 \\ \bar{w}L^2/30 \\ 0 \\ 7\bar{w}L/20 \\ -\bar{w}L^2/20 \end{Bmatrix}$
Equivalent nodal loading definition for a transverse point load		
Beam Element Formulation	Frame Element Formulation	
	$f_{eq,PL}^e = \bar{w} \begin{Bmatrix} 1 - \frac{3a^2}{L^2} + \frac{2a^3}{L^3} \\ \frac{a^3}{L^2} - \frac{2a^2}{L} + a \\ \frac{3a^2}{L^2} - \frac{2a^3}{L^3} \\ \frac{a^3}{L^2} - \frac{a^2}{L} \end{Bmatrix}$	$f_{eq,PL}^e = \bar{w} \begin{Bmatrix} 0 \\ 1 - 3(a^2/L^2) + 2(a^3/L^3) \\ (a^3/L^2) - 2(a^2/L) + a \\ 0 \\ 3(a^2/L^2) - 2(a^3/L^3) \\ (a^3/L^2) - (a^2/L) \end{Bmatrix}$
Specific, equivalent nodal loading definition for a mid-span point load, where $a = L/2$		
	$f_{eq,PL}^e = \begin{Bmatrix} \bar{w}/2 \\ \bar{w}L/8 \\ \bar{w}/2 \\ -\bar{w}L/8 \end{Bmatrix}$	$f_{eq,PL\{6 \times 1\}}^e = \begin{Bmatrix} 0 \\ \bar{w}/2 \\ \bar{w}L/8 \\ 0 \\ \bar{w}/2 \\ -\bar{w}L/8 \end{Bmatrix}$
Equivalent nodal loading definition for a Distributed Axial Load		
Bar Element Formulation	Frame Element Formulation	
	$f_{eq,axial}^e = \bar{p} \begin{Bmatrix} L/2 \\ L/2 \end{Bmatrix}$	$f_{eq,axial}^e = \bar{p} \begin{Bmatrix} L/2 \\ 0 \\ 0 \\ L/2 \\ 0 \\ 0 \end{Bmatrix}$
Equivalent nodal loading definition for a Concentrated Axial Load		
Bar Element Formulation	Frame Element Formulation	
	$f_{eq,axial PL}^e = \bar{p} \begin{Bmatrix} 1 - a/L \\ a/L \end{Bmatrix}$	$f_{eq,axial PL}^e = \bar{p} \begin{Bmatrix} 1 - a/L \\ 0 \\ 0 \\ a/L \\ 0 \\ 0 \end{Bmatrix}$

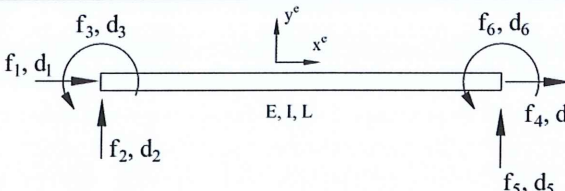
Problem 1:

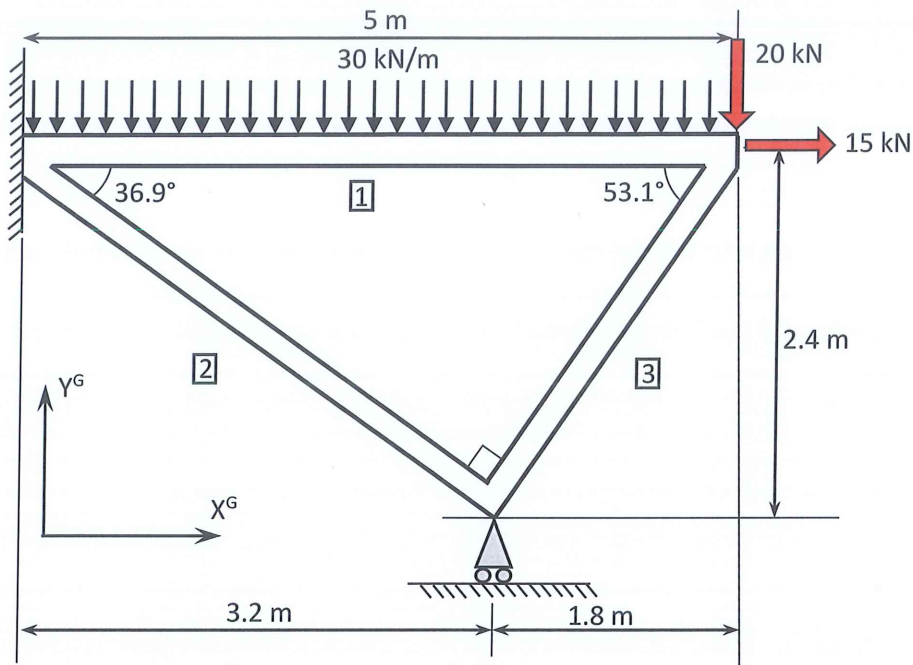
A steel structure is modelled as three frame elements, as shown on the next page. The support at the left edge of the structure is considered to be a fixed support and the support under the bottom of the structure is a roller support that can provide a vertical reaction force but cannot provide any horizontal force or moment reaction. The structure itself is considered mass-less relative to the loads it carries.

A downwards uniformly distributed load with intensity of 30 kN/m is applied to the top of element 1. A downwards concentrated load of 20 kN and a horizontal load of 15 kN acting to the right are applied at the top right-hand corner of the structure, as shown on the next page.

Assume that all elements are frame elements, made of steel, and that each element has the same cross-sectional area and second moment of area, with values given with the sketch on the next page. The length of the elements differ (as given on the next page). **[21 Total Marks]**

- Draw a sketch of the structure showing the global degrees of freedom $\mathbf{q}$ for the model. **[2 Marks]**
- Draw a free-body diagram for each element, showing the element degrees of freedom in global co-ordinates as well as the orientation of your element co-ordinate system (including stating the transformation angle, α , for each element). **[2 Marks]**
- Write out the assembly matrices $\mathbf{A}^e$ for each element **[2 Marks]**
- Write out the element equivalent forcing terms for element 1 ($\mathbf{f}_{eq}^1$) to represent the Uniformly Distributed Load. Present the overall global forcing vector, $\mathbf{Q}$, that includes both the UDL and the point loads. **[2 Marks]**
- Solve for the overall deflections of the system, $\mathbf{q}$, and present your result. **[1 Mark]**
- Sketch (by hand) the original and deflected shape of the structure, indicating the direction and magnitude of all deflection components (displacements and rotations). The sketch does not need to be to scale, but each component of deflection must be labelled. **[3 Marks]**
- Provide the reaction components (forces and moments) at all supports, parallel and perpendicular to the supports. Provide a sketch of these reaction loads, showing the magnitude and direction of all reaction components. **[3 Marks]**
- Determine the maximum bending moment magnitude (positive or negative) induced within element 2 (M_{max}^2) and state where it occurs within the element. **[1 Mark]**
- Determine the axial normal strain in element 3 (from axial loads only, ignoring any bending effects), and indicate whether it is tensile or compressive. **[1 Mark]**
- Present the element displacement vector (in element co-ordinates) for element 2 and determine the total deflection (in the X^G and Y^G directions) of a point located at a distance of 3m (along the element) from the top left of element 2. The shape functions (applied to the frame element numbering system) are given below. Include your working, including presenting the values for both $u(x)$ and $v(x)$ at this location. **[4 Marks]**

List of the shape functions for a frame element.			
$\psi_1(x) = \left(1 - \frac{x}{L}\right); \psi_2(x) = \frac{x}{L}$			
$N_1(x) = 1 - \frac{3x^2}{L^2} + \frac{2x^3}{L^3}$	$N_2(x) = \frac{x^3}{L^2} - \frac{2x^2}{L} + x$	$N_3(x) = \frac{3x^2}{L^2} - \frac{2x^3}{L^3}$	$N_4(x) = \frac{x^3}{L^2} - \frac{x^2}{L}$
		$u(x) = \psi_1(x)d_1 + \psi_2(x)d_2$ $v(x) = \mathbf{N} \mathbf{d}^e = \begin{pmatrix} N_1(x) & N_2(x) & N_3(x) & N_4(x) \end{pmatrix} \begin{pmatrix} d_1 \\ d_2 \\ d_3 \\ d_4 \end{pmatrix}$	



All elements are FRAME elements

Elastic Modulus: $E^1 = E^2 = E^3 = 200 \text{ GPa}$

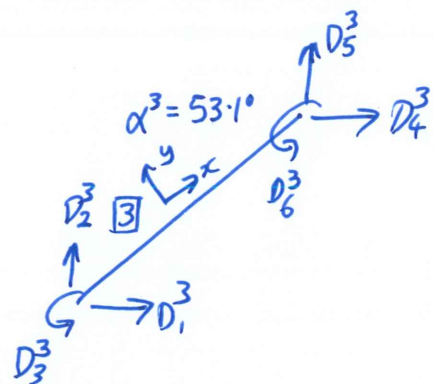
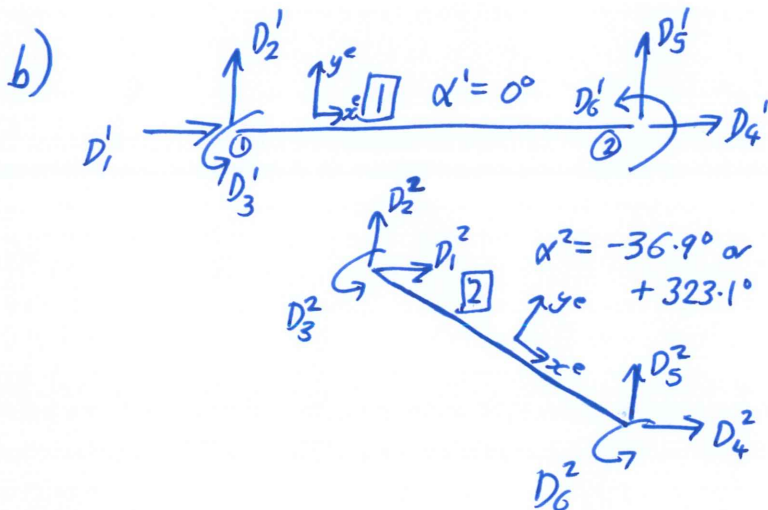
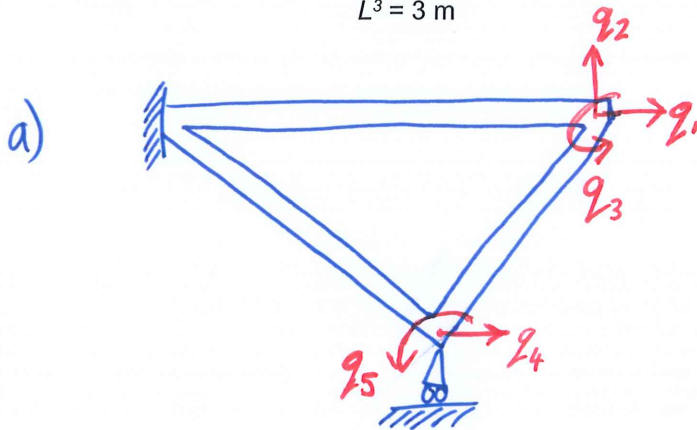
Second Moment of Area: $I^1 = I^2 = I^3 = 5 \times 10^{-6} \text{ m}^4$

Cross-sectional Area: $A^1 = A^2 = A^3 = 3 \times 10^{-4} \text{ m}^2$

Lengths: $L^1 = 5 \text{ m}$

$L^2 = 4 \text{ m}$

$L^3 = 3 \text{ m}$



$$c) \quad A^1 = \begin{matrix} & D_1^1 & D_2^1 & D_3^1 & D_4^1 & D_5^1 & D_6^1 \\ \begin{matrix} q_1 \\ q_2 \\ q_3 \\ q_4 \\ q_5 \end{matrix} & \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

$$A^2 = \begin{matrix} & D_1^2 & D_2^2 & D_3^2 & D_4^2 & D_5^2 & D_6^2 \\ \begin{matrix} q_1 \\ q_2 \\ q_3 \\ q_4 \\ q_5 \end{matrix} & \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \end{matrix}$$

$$A^3 = \begin{matrix} & D_1^3 & D_2^3 & D_3^3 & D_4^3 & D_5^3 & D_6^3 \\ \begin{matrix} q_1 \\ q_2 \\ q_3 \\ q_4 \\ q_5 \end{matrix} & \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

$$d) \quad f_{eq, uoL}^1 = \bar{w} \begin{Bmatrix} 0 \\ L/2 \\ (L')^2/12 \\ 0 \\ L/2 \\ -(L')^2/12 \end{Bmatrix} = (-30,000) \begin{Bmatrix} 0 \\ 5/2 \\ 5^2/12 \\ 0 \\ 5/2 \\ -(5^2)/12 \end{Bmatrix} = \begin{Bmatrix} 0 \text{ N} \\ -75,000 \text{ N} \\ -62,500 \text{ Nm} \\ 0 \text{ N} \\ -75,000 \text{ N} \\ +62,500 \text{ Nm} \end{Bmatrix}$$

$$F_{eq, uoL}^1 = (\underline{1})^T f_{eq, uoL}^1 = \begin{Bmatrix} 0 \text{ N} \\ -75,000 \text{ N} \\ -62,500 \text{ Nm} \\ 0 \text{ N} \\ -75,000 \text{ N} \\ +62,500 \text{ Nm} \end{Bmatrix}$$

In this case, as $\alpha' = 0^\circ$

$\underline{1} = I_{6 \times 6}$ identity matrix

so $F_{eq, uoL}^1 = f_{eq, uoL}^1$

4 non-zero values in $F_{eq, uoL}^1$

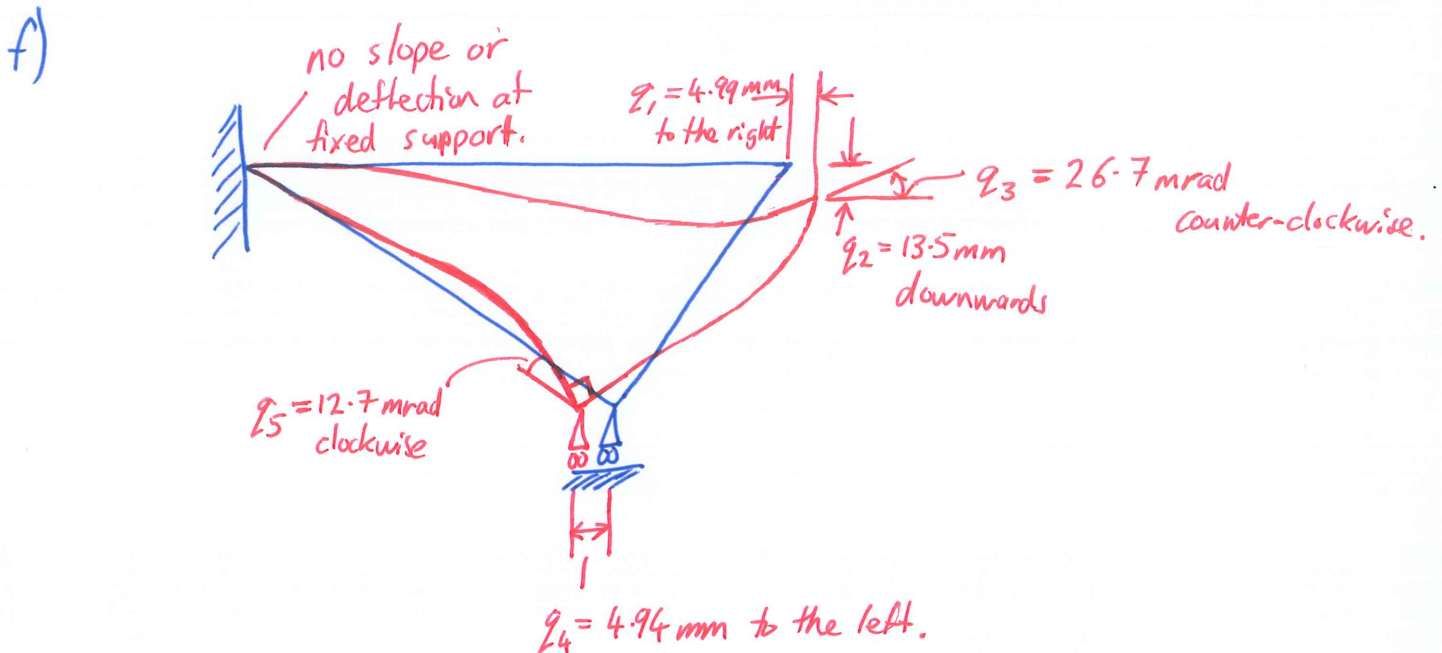
$$Q_{eq, uoL}^1 = A^1 F_{eq, uoL}^1 = \begin{Bmatrix} 0 \text{ N} \\ -75,000 \text{ N} \\ +62,500 \text{ Nm} \\ 0 \text{ N} \\ 0 \text{ Nm} \end{Bmatrix}$$

2 non-zero values in $Q_{eq, uoL}^1$

d) continued. $Q_{\text{nodal}} = \begin{Bmatrix} 15,000 \text{ N} \\ -20,000 \text{ N} \\ 0 \text{ Nm} \\ 0 \text{ N} \\ 0 \text{ Nm} \end{Bmatrix}$

$$Q_{\text{total}} = Q_{\text{nodal}} + Q_{\text{eq, nodal}} = \begin{Bmatrix} 15,000 \text{ N} \\ -20,000 \text{ N} \\ 0 \text{ Nm} \\ 0 \text{ N} \\ 0 \text{ Nm} \end{Bmatrix} + \begin{Bmatrix} 0 \text{ N} \\ -75,000 \text{ N} \\ +62,500 \text{ Nm} \\ 0 \text{ N} \\ 0 \text{ Nm} \end{Bmatrix} = \begin{Bmatrix} +15,000 \text{ N} \\ -95,000 \text{ N} \\ +62,500 \text{ Nm} \\ 0 \text{ N} \\ 0 \text{ Nm} \end{Bmatrix}$$

e) $q = \text{np.linalg.solve}(KG, Q_{\text{total}}) = \begin{Bmatrix} 0.0049933 \text{ m} \\ -0.01351348 \text{ m} \\ 0.02672677 \text{ rad} \\ -0.00494613 \text{ m} \\ -0.01270269 \text{ rad} \end{Bmatrix} = \begin{Bmatrix} 4.99 \text{ mm} \\ -13.5 \text{ mm} \\ 26.7 \text{ mrad} \\ -4.94 \text{ mm} \\ -12.7 \text{ mrad} \end{Bmatrix}$



g)

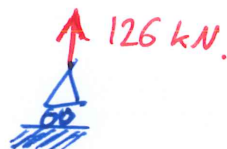
$$F'_{1-3} + F_{1-3} - F_{\text{eq, nodal}, 1-3}$$

$$\begin{Bmatrix} -59.9196 \\ 7.7117 \\ 13.933 \end{Bmatrix} + \begin{Bmatrix} 44.919 \\ -38.987 \\ -5.2376 \end{Bmatrix} - \begin{Bmatrix} 0 \\ -75.0 \\ -62.5 \end{Bmatrix}$$

$$= \begin{Bmatrix} -15.0 \text{ kN} \\ 43.7 \text{ kN} \\ 71.2 \text{ kNm} \end{Bmatrix}$$

$$F_{4-6}^2 + F_{1-3}^3 = \begin{Bmatrix} -44.919 \\ 38.987 \\ -11.589 \end{Bmatrix} + \begin{Bmatrix} 44.919 \\ 87.288 \\ 11.589 \end{Bmatrix}$$

$$= \begin{Bmatrix} 0 \text{ kN} \\ 126.275 \text{ kN} \\ 0 \text{ kNm} \end{Bmatrix}$$



h)

$$F^2 = \begin{Bmatrix} 44.9196 \text{ kN} \\ -38.9870 \text{ kN} \\ -5.23769 \text{ kNm} \\ -44.9196 \text{ kN} \\ 38.9870 \text{ kN} \\ -11.5890 \text{ kNm} \end{Bmatrix}$$

Element [2] carries a constant shear force, so the moment varies linearly from the value at node ① to the value at node ③.

Therefore, the maximum moment will be 11.6 kNm

and it occurs at node ②, which is at the right hand end.

i)

$$d^3 = \begin{Bmatrix} -2.96975 \text{ mm} \\ 3.95534 \text{ mm} \\ -12.70269 \text{ mrad} \\ -7.80844 \text{ mm} \\ -12.10683 \text{ mm} \\ 26.72677 \text{ mrad} \end{Bmatrix}$$

$$\epsilon^3 = \frac{d_4^3 - d_1^3}{L^3}$$

$$\epsilon^3 = \frac{(-7.80844 \times 10^{-3}) - (-2.96975 \times 10^{-3})}{3}$$

$$= -1,613 \mu\epsilon$$

as this strain is negative, the strain is compressive.

$$j) \quad d^2 = \begin{Bmatrix} 0 \text{ mm} \\ 0 \text{ mm} \\ 0 \text{ mrad} \\ -3.9553 \text{ mm} \\ -2.9697 \text{ mm} \\ -12.7027 \text{ mrad} \end{Bmatrix}$$

$$u(x) = \psi_1(x) d_1 + \psi_2(x) d_4$$

$$v(x) = \cancel{\mathcal{N}_1(x) d_2} + \cancel{\mathcal{N}_2(x) d_3} + \mathcal{N}_3(x) d_5 + \mathcal{N}_4(x) d_6$$

$$\psi_2(x) = \frac{x}{L} = \frac{3}{4} = 0.75$$

$$u(x) = (0.75)(-3.9553) = -2.9665 \text{ mm.}$$

$$\mathcal{N}_3(x) = \frac{3x^2}{L^2} - \frac{2x^3}{L^3} = \frac{3(3)^2}{4^2} - \frac{2(3)^3}{4^3} = 1.6875 - 0.84375 = 0.84375.$$

$$\mathcal{N}_4(x) = \frac{x^3}{L^2} - \frac{x^2}{L} = \frac{3^3}{4^2} - \frac{3^2}{4} = 1.6875 - 2.25 = -0.5625.$$

$$v(x) = \mathcal{N}_3(x) d_5 + \mathcal{N}_4(x) d_6 = (0.84375)(-2.9697 \times 10^{-3}) + (-0.5625)(-12.7027 \times 10^{-3})$$

$$v(x) = 4.639 \text{ mm.}$$

$$\delta X^G = u(x) \cos(\alpha) - v(x) \sin(\alpha)$$

$$= -2.9665 \cos(-36.9^\circ) - (4.639) \sin(-36.9^\circ)$$

$$= 0.4105 \text{ mm}$$

$$\delta Y^G = u(x) \sin(\alpha) + v(x) \cos(\alpha)$$

$$= (-2.9665) \sin(-36.9^\circ) + (4.639) \cos(-36.9^\circ)$$

$$= 5.491 \text{ mm.}$$

Problem 2: A photograph of the bike shed on Creyke Road near the entrance to Rātā is shown on the next page. Structural frames are used to support a roof that keeps the bikes dry during adverse weather.

We will simplify the problem by using symmetry and only model the left half of the structure. The boundary condition at the bottom left corner of the half-structure is a fixed support. The boundary conditions that we will apply at the middle of the structure (at the plane of symmetry) are rollers which prevent rotation and horizontal translation, but do not provide any vertical reaction force.

The half-structure that we will model is made up of two frame elements. The structure supports a distributed roof load of 1 kPa, which can be transformed into a Uniformly Distributed Load (UDL) acting directly downwards on element 2.

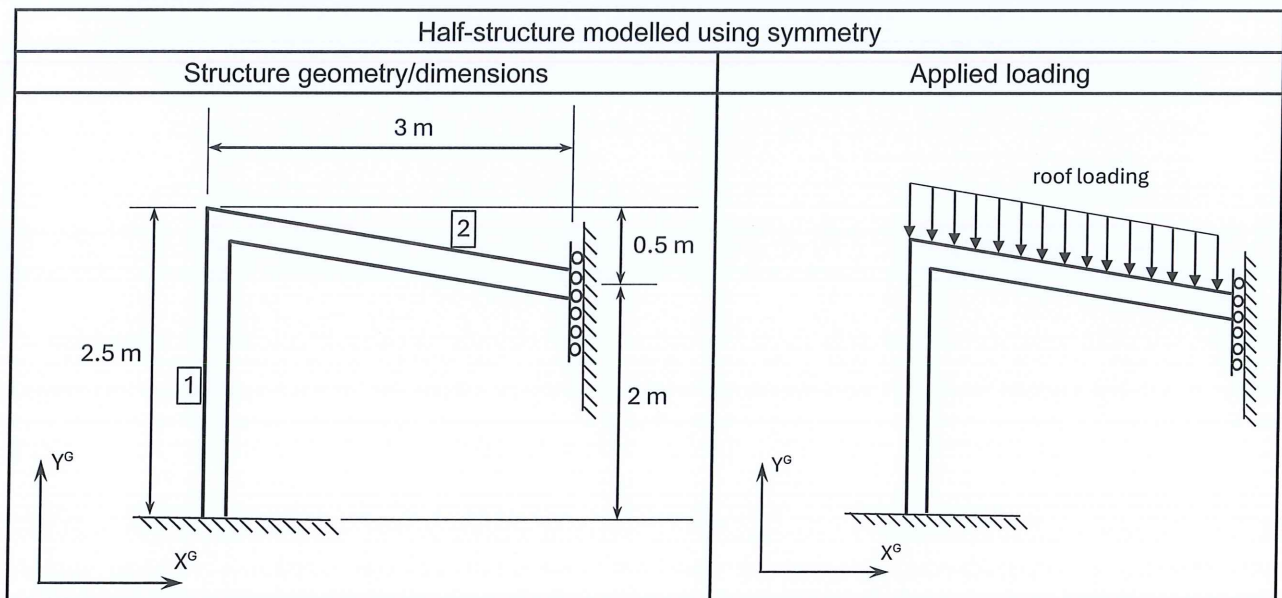
Assume that all elements are frame elements, made of steel, and that each element has the same cross-sectional area and second moment of area, with values given on the next page. The length of the elements differ and can be determined from the diagram. The structure itself is considered mass-less relative to the loads it carries, so the only applied load is the UDL. **[19 Total Marks]**

- a. Draw a sketch of the structure showing the global degrees of freedom $\mathbf{q}$. **[2 Marks]**
- b. Draw the free body diagram of each element, showing the element degrees of freedom in global coordinates and the orientation of your element co-ordinate system (including stating the transformation angle, α , for each element). **[2 Marks]**
- c. Write out the assembly matrices $\mathbf{A}^e$ for each element **[2 Marks]**
- d. Assume that the frame being modelled supports a section of roof (the distance into the page) of 4m, determine the loading on element 2 (in kN/m) from the given value of roof loading of 1 kPa.

Write out the element equivalent forcing terms for element 2 ($\mathbf{f}_{eq}^2$) to represent the roof loading. Present both the axial and shear components of the distributed load (these will be a 6x1 vectors), as well as the final overall global forcing vector, $\mathbf{Q}$.

If you cannot calculate the intensity of the uniformly distributed load that represents the roof loading, use a magnitude of $\bar{w}$ of 7.5 kN/m.

***Note:** The formulae on the second page of this test may be a useful reference. If you are unsure how to break up the shear and axial components of the distributed load, just apply the roof load as a pure shear load and continue with the question.* **[4 Marks]**
- e. Solve for the overall deflections of the system, $\mathbf{q}$, and present your result. **[1 Mark]**
- f. Clearly sketch the overall deflected shape of the system.
→ In your sketch, show all the deflections and rotations that were calculated and label the magnitudes. The sketch does not need to be to scale, but each component must be labelled. **[3 Marks]**
- g. Provide the support reaction forces and moments at both support points, parallel and perpendicular to the supports. Provide a sketch for each case, showing the magnitude and direction of all reaction components. **[3 Marks]**
- h. If the maximum bending moment within element 1 is restricted to be no more than 25 kNm, what is the maximum intensity of the roof loading (in kPa) that can be applied to this structure. **[2 Marks]**



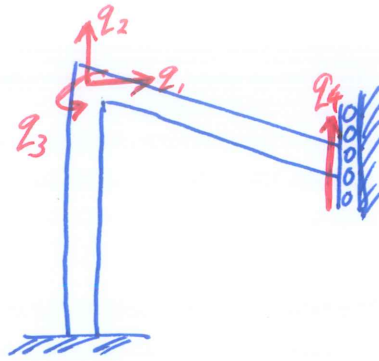
All elements are FRAME elements

Elastic Modulus: $E^1 = E^2 = 200 \text{ GPa}$

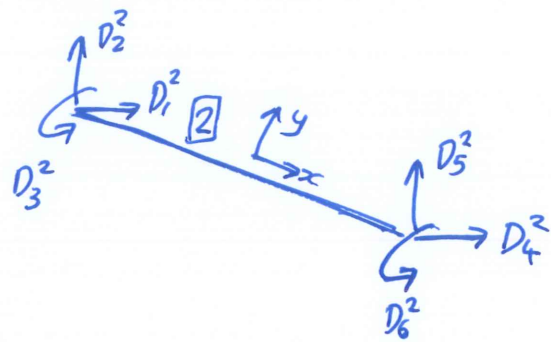
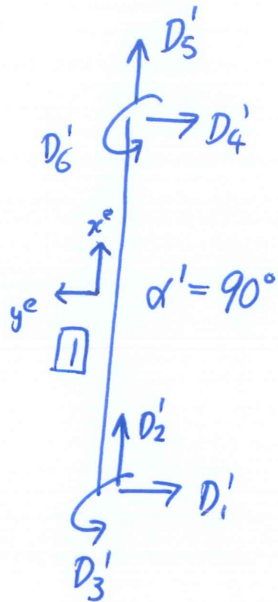
Second Moment of Area: $I^1 = I^2 = 5 \times 10^{-6} \text{ m}^4$

Cross-sectional Area: $A^1 = A^2 = 3 \times 10^{-4} \text{ m}^2$

a)



b)



$$\alpha^2 = -\tan^{-1}\left(\frac{0.5}{3}\right)$$

$$\alpha^2 = -9.462^\circ$$

NOTE: $L^3 = \sqrt{3^2 + (0.5)^2}$
 $= \sqrt{9.25}$

$$L^3 = 3.0414 \text{ m.}$$

c)

$$A' = \begin{matrix} & D_1' & D_2' & D_3' & D_4' & D_5' & D_6' \\ \begin{matrix} q_1 \\ q_2 \\ q_3 \\ q_4 \end{matrix} & \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

$$A^2 = \begin{matrix} & D_1^2 & D_2^2 & D_3^2 & D_4^2 & D_5^2 & D_6^2 \\ \begin{matrix} q_1 \\ q_2 \\ q_3 \\ q_4 \end{matrix} & \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

d) UDL intensity = $(1 \text{ kPa})(4\text{m}) = 4 \text{ kN/m}$.

Break up this intensity into a shear and axial component.



$\bar{w} = -4000 \cos(9.462^\circ)$
 $\bar{w}_{\text{shear}} = -3,945.6 \text{ kN/m}$
 $\bar{p}_{\text{axial}} = 4000 \sin(9.462^\circ)$
 $= 657.6 \text{ kN/m}$

$$f_{\text{eq, shear}}^2 = \bar{w} \begin{Bmatrix} 0 \\ L^2/2 \\ (L^3)/12 \\ 0 \\ L^2/2 \\ -(L^3)/12 \end{Bmatrix} = (-3,945.6) \begin{Bmatrix} 0 \\ 3.0414/2 \\ 3.0414^2/12 \\ 0 \\ 3.0414/2 \\ -(3.0414)^2/12 \end{Bmatrix} = \begin{Bmatrix} 0 \text{ N} \\ -6,000 \text{ N} \\ -3,041.4 \text{ Nm} \\ 0 \text{ N} \\ -6,000 \text{ N} \\ +3,041.4 \text{ Nm} \end{Bmatrix}$$

$$F_{\text{eq, shear}}^2 = (\mathbf{1}^3)^T f_{\text{eq, shear}}^2 = \begin{Bmatrix} -986.4 \text{ N} \\ -5918.4 \text{ N} \\ -3,041.4 \text{ Nm} \\ -986.4 \text{ N} \\ -5918.4 \text{ N} \\ +3,041.4 \text{ Nm} \end{Bmatrix} \quad \sim 6 \text{ non-zero values.}$$

$$Q_{\text{eq, shear}}^2 = \mathbf{A}^2 F_{\text{eq, shear}}^2 = \begin{Bmatrix} -986.4 \text{ N} \\ -5918.4 \text{ N} \\ -3,041.4 \text{ Nm} \\ -5918.4 \text{ N} \end{Bmatrix} \quad \sim 4 \text{ non-zero values}$$

$$f_{\text{eq, axial}}^2 = \bar{p} \begin{Bmatrix} L^2/2 \\ 0 \\ 0 \\ L^2/2 \\ 0 \\ 0 \end{Bmatrix} = (657.6) \begin{Bmatrix} 3.0414/2 \\ 0 \\ 0 \\ 3.0414/2 \\ 0 \\ 0 \end{Bmatrix} = \begin{Bmatrix} 1,000 \text{ N} \\ 0 \text{ N} \\ 0 \text{ Nm} \\ 1,000 \text{ N} \\ 0 \text{ N} \\ 0 \text{ Nm} \end{Bmatrix}$$

$$F_{\text{eq, axial}}^2 = (\mathbf{1}^2)^T f_{\text{eq, axial}}^2 = \begin{Bmatrix} 986.4 \text{ N} \\ -164.4 \text{ N} \\ 0 \text{ Nm} \\ 986.4 \text{ N} \\ -164.4 \text{ N} \\ 0 \text{ Nm} \end{Bmatrix}$$

$$Q_{\text{eq, axial}}^2 = \mathbf{A}^2 F_{\text{eq, axial}}^2 = \begin{Bmatrix} 986.4 \text{ N} \\ -164.4 \text{ N} \\ 0 \text{ Nm} \\ -164.4 \text{ N} \end{Bmatrix}$$

d) continued.

$$Q_{total} = Q_{eq, shear}^2 + Q_{eq, axial}^2$$

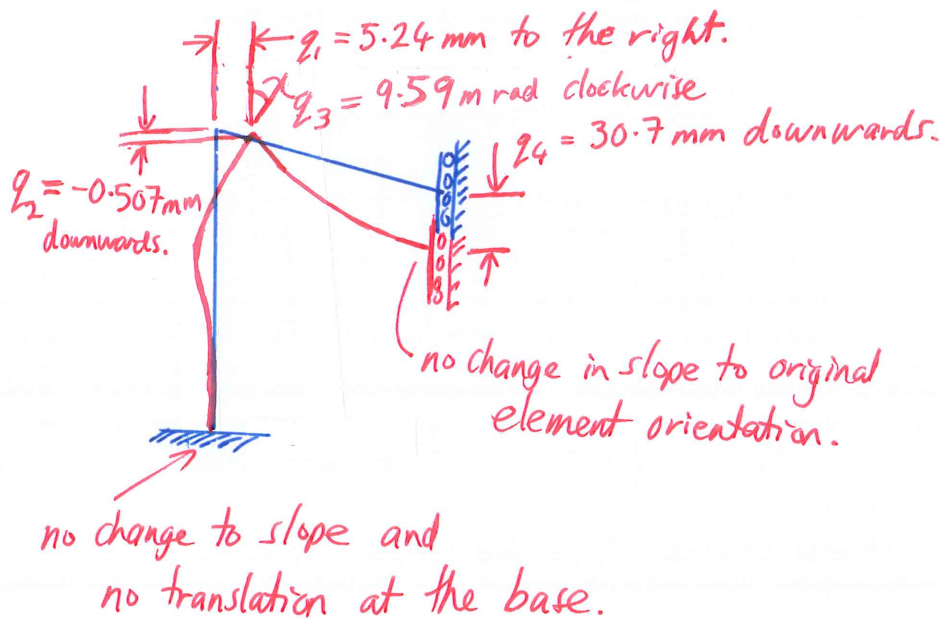
$$= \begin{Bmatrix} -986.4 \text{ N} \\ -5918.4 \text{ N} \\ -3,041.4 \text{ Nm} \\ -5,918.4 \text{ N} \end{Bmatrix} + \begin{Bmatrix} 986.4 \text{ N} \\ -164.4 \text{ N} \\ 0 \text{ Nm} \\ -164.4 \text{ N} \end{Bmatrix}$$

$$Q_{total} = \begin{Bmatrix} 0 \text{ N} \\ -6,082.8 \text{ N} \\ -3,041.4 \text{ Nm} \\ -6,082.8 \text{ N} \end{Bmatrix}$$

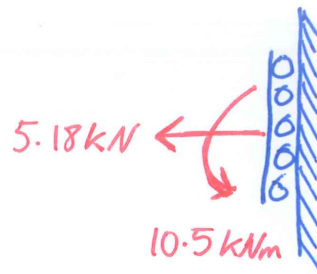
e) $q = \text{numpy.linalg.solve}(KG, Q_{total})$

$$q = \begin{Bmatrix} 0.00524294 \text{ m} \\ -0.0005069 \text{ m} \\ -0.00958794 \text{ rad} \\ -0.03069806 \text{ m} \end{Bmatrix} = \begin{Bmatrix} 5.243 \text{ mm} \\ -0.5069 \text{ mm} \\ -9.5879 \text{ mrad} \\ -30.698 \text{ mm} \end{Bmatrix}$$

f)

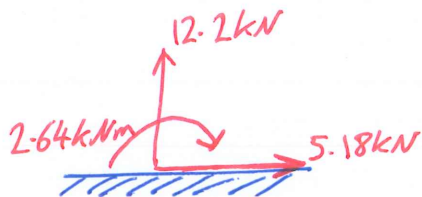


9)



$$F_{4-6}^2 + (-F_{eq, shear, 4-6}^2) + (-F_{eq, axial, 4-6}^2)$$

$$\begin{Bmatrix} -5.1779 \text{ kN} \\ -6.0828 \text{ kN} \\ 13.5711 \text{ kNm} \end{Bmatrix} - \begin{Bmatrix} -0.9864 \text{ kN} \\ -5.9184 \text{ kN} \\ 3.0414 \text{ kNm} \end{Bmatrix} - \begin{Bmatrix} 0.9864 \text{ kN} \\ -0.1644 \text{ kN} \\ 0 \text{ kNm} \end{Bmatrix}$$



$$= \begin{Bmatrix} -5.1779 \text{ kN} \\ 0 \text{ kN} \\ 10.5297 \text{ kNm} \end{Bmatrix}$$

$$F_{1-3}^1 = \begin{Bmatrix} 5.1779 \text{ kN} \\ 12.1655 \text{ kN} \\ -2.6371 \text{ kNm} \end{Bmatrix}$$

h) The current maximum bending moment in element [I] is a magnitude of 10.3 kNm at the top node.

This is a linear elastic analysis, and the roof loading is the only applied load.

Therefore, applied load and the maximum bending moment in element [I] will be linearly related.

$$\text{Max roof loading} = \frac{M_{\max}}{M_{@1 \text{ kPa}}} \times 1 \text{ kPa} = \frac{25.0}{10.30749} \times 1.0 = 2.425 \text{ kPa.}$$

Max roof loading = 2.43 kPa to limit the moment in element [I] to 25 kNm